(00) (5 分)

运动方程对称性、正则坐标成积量纲、变换灵活、易于呈现与量子力学的亲密关系

说出两点即可

(01) (8 分)

$$\frac{d\vec{L}}{dt}=\vec{\tau}_o^{(\text{ext})}=\vec{r}_c\times\vec{F}^{(\text{ext})},\qquad \frac{d\vec{S}}{dt}=\vec{\tau}_s^{(\text{ext})}=\sum_i\vec{r}_i'\times\vec{f}_i^{(\text{ext})},$$

有心力场中的质点系，取力心为坐标原点，外力矩 $\sum_i\vec{r}_i\times f_i\hat{r}_i=0$，故体系总角动量守恒

但质点系受到的“轨道外力矩”和“自旋外力矩”可以不为 0，从而轨道角动量和自旋角动量的改变量刚好相反。称为轨道角动量和自旋角动量可以互相转换。

(02) (6 分)

(a), (c) 正确 选对一个的 3 分，选错一个扣 3 分

(03) (6 分)

(a), (d) (f) 正确 选对一个的 2 分，选错一个扣 2 分，

(4) (6 分)

$$\dot{\xi}=\frac{\partial H}{\partial p_\xi}=\frac{p_\xi}{m\left(\xi^2+\eta^2\right)},\quad \dot{\eta}=\frac{\partial H}{\partial p_\eta}=\frac{p_\eta}{m(\xi^2+\eta^2)},\quad \dot{\phi}=\frac{\partial H}{\partial p_\phi}=\frac{p_\phi}{m\,\xi^2\,\eta^2},$$

$$L=p_\xi\dot{\xi}+p_\eta\dot{\eta}+p_\phi\dot{\phi}-H=\frac{m}{2}\left(\xi^2+\eta^2\right)\left(\dot{\xi}^2+\dot{\eta}^2\right)+\frac{m}{2}\xi^2\eta^2\dot{\phi}^2$$

(05) (12 分)

沿 x 方向平移：$\begin{cases}x'=x+\epsilon\\ y'=y\\ z'=z\end{cases}\Longrightarrow\begin{cases}\delta x=\epsilon\\ \delta y=0\\ \delta z=0\end{cases}\quad\begin{cases}\delta\dot{x}=0\\ \delta\dot{y}=0\\ \delta\dot{z}=0\end{cases}$

$$\delta L=\frac{\partial L}{\partial x}\delta x=(e\,B/2)\,\dot{y}\,\epsilon=\frac{d}{dt}\left(\frac{e\,B\,y}{2}\right)\epsilon,\qquad \mathcal{F}(q,\,t)=\frac{e\,B\,y}{2}$$

守恒量：$\mu_x=\frac{\partial L}{\partial\dot{x}}-\mathcal{F}=m\,\dot{x}-\frac{e\,B\,y}{2}-\frac{e\,B\,y}{2}=m\,\dot{x}-e\,B\,y$

类似可求得：$\mu_y=\left(m\,\dot{y}+\frac{e\,B}{c}x\right)$,

类似可求得：$\mu_z=m\,\dot{z}=p_3$，　　对应于可遗坐标 z

(06) (6 分)

$$[x,\,L_x]=\left[x,\,y\,p_z-z\,p_y\right]=0,\quad \left[x,\,L_y\right]=[x,\,z\,p_x-x\,p_z]=z,\quad [x,\,L_z]=\left[x,\,x\,p_y-y\,p_x\right]=-y$$

$$\left[x,\,\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{L}}\right]=n_y\,z-n_z\,y\qquad\Longrightarrow\quad\left[\vec{r},\,\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{L}}\right]=\hat{\boldsymbol{n}}\times\vec{r}$$

(07) (12 分)

完整的拉格朗日量 $L=\frac{m}{2}\left(\dot{\rho}^2+\rho^2\dot{\phi}^2+\dot{z}^2\right)-m\,g\,z$

代入约束条件可得约化的拉格朗日量：

$$L=\frac{m}{2}\left(\dot{\rho}^2+\rho^2\omega^2+4\,\alpha^2\rho^2\dot{\rho}^2\right)-m\,g\,\alpha\,\rho^2=\frac{m}{2}\left(4\,\alpha^2\rho^2+1\right)\dot{\rho}^2$$

$$H=\frac{\partial L}{\partial\dot{\rho}}\dot{\rho}-L=\frac{m}{2}\left(4\,\alpha^2\rho^2+1\right)\dot{\rho}^2$$

$\frac{dH}{dt}=-\frac{\partial L}{\partial t}=0$ 故 $H=\frac{m}{2}\left(4\,\alpha^2\rho^2+1\right)\dot{\rho}^2$ 是运动常数

该运动常数和时间平移对称性相关。

(08) (8 分)

平面曲线，取极坐标：$(r,\,\theta)$,

$$ds = \sqrt{(dr)^2 + (r\,d\theta)^2} = \sqrt{\dot r^2 + r^2\dot\theta^2}\,d\beta, \quad 其中\ \dot r = \frac{dr}{d\beta}, \quad \dot\theta = \frac{d\theta}{d\beta}$$

$$I = \int_A^B \sqrt{2m(E-V)}\sqrt{\dot r^2 + r^2\dot\theta^2}\,d\beta \quad 驻值条件：\frac{d}{d\beta}\frac{\partial f}{\partial\dot\theta} + \frac{\partial f}{\partial\theta} = 0 \implies \frac{\partial f}{\partial\dot\theta} = \text{const}$$

$$\frac{\sqrt{2m(E-V)}}{\sqrt{\dot r^2 + r^2\dot\theta^2}}\,r^2\dot\theta = c \quad 令\ \beta = \theta \quad \implies \frac{dr}{d\theta} = \pm\frac{r^2}{c}\left[2m(E-V) - \frac{c^2}{r^2}\right]^{1/2}$$

(09) (22 分)

力与速度有关，应该利用广义势 $U(r,\dot r)$，

(a) 广义力 F 作用下的拉格朗日方程：$\frac{d}{dt}\frac{\partial L}{\partial\dot r} - \frac{\partial L}{\partial r} = Q_r = F$

(b) 若可以构造广义势 $U(r,\dot r)$ 并将之归入拉格朗日量，新拉格朗日方程为

$$\frac{d}{dt}\frac{\partial[L - U(r,\dot r)]}{\partial\dot r} - \frac{\partial[L - U(r,\dot r)]}{\partial r} = 0 \implies \frac{d}{dt}\frac{\partial L}{\partial\dot r} - \frac{\partial L}{\partial r} = \frac{d}{dt}\frac{\partial U(r,\dot r)}{\partial\dot r} - \frac{\partial U(r,\dot r)}{\partial r}$$

(c) 比较两个方程可得对应的广义力为

$$Q_r = \frac{d}{dt}\frac{\partial U}{\partial\dot r} - \frac{\partial U}{\partial r} = F = \frac{\alpha}{r^2}\left(1 - \frac{\dot r^2 - 2r\ddot r}{c^2}\right) = \frac{\alpha}{r^2} - \frac{\alpha\dot r^2}{c^2 r^2} + \frac{2\alpha\ddot r}{r c^2}$$

(d) 设 $U(r,\dot r) = \alpha f(r)\dot r^2 + \alpha g(r)$

代入得：$2\alpha f(r)\ddot r + 2\alpha f'(r)\dot r^2 - \alpha f'(r)\dot r^2 - \alpha g'(r) = \frac{\alpha}{r^2} - \frac{\alpha\dot r^2}{c^2 r^2} + \frac{2\alpha\ddot r}{r c^2}$

比较 $\dot r$ 和 $\ddot r$ 的“系数”可得：$f(r) = \frac{1}{c^2 r}$, $\quad g'(r) = -\frac{1}{r^2} \implies g(r) = \frac{1}{r} + c$

$$U(r,\dot r) = \frac{\alpha\dot r^2}{c^2 r} + \frac{\alpha}{r} + c,$$

拉格朗日：$L = \frac{m}{2}\left(\dot r^2 + r^2\dot\phi^2\right) - \frac{\alpha}{r}\left(\frac{\dot r^2}{c^2} + 1\right)$

$$p_r = \frac{\partial L}{\partial\dot r} = m\dot r - \frac{2\alpha}{c^2}\frac{\dot r}{r}, \quad p_\phi = m r^2\dot\phi \implies \dot r = \frac{p_r}{m - 2\alpha/(c^2 r)}; \quad \dot\phi = \frac{p_\phi}{m r^2}$$

$$H = p_r\dot r + p_\phi\dot\phi - L = \frac{p_r^2}{m - 2\alpha/(c^2 r)} + \frac{p_\phi^2}{m r^2} - \frac{m}{2}\frac{p_r^2}{[m - 2\alpha/(c^2 r)]^2} - \frac{m}{2}r^2\frac{p_\phi^2}{m^2 r^4} + \frac{\alpha}{r}\left(\frac{p_r^2}{c^2[m - 2\alpha/(c^2 r)]^2} + 1\right)$$

$$H = \frac{p_r^2}{2\left(m - \frac{2\alpha}{c^2 r}\right)} + \frac{p_\phi^2}{2 m r^2} + \frac{\alpha}{r}$$

(10) (20 分)

对 $k = 1, 2, \ldots, D$，$\frac{d}{d\beta}\frac{\partial\mathcal{L}(q,\dot q)}{\partial q_k'} - \frac{\partial\mathcal{L}(q,\dot q)}{\partial q_k} = 0$

$$\frac{\partial\mathcal{L}(q,\dot q)}{\partial q_k'} = \frac{\partial q_0' L\left(q_{[0]}, \frac{q_{[0]}'}{q_0'}, t\right)}{\partial q_k'} = q_0'\frac{\partial L\left(q_{[0]}, \frac{q_{[0]}'}{q_0'}, t\right)}{\partial q_k'} = \frac{\partial L(q_{[0]}, \dot q_{[0]}, t)}{\partial\dot q_k}$$

$$\frac{\partial\mathcal{L}(q,\dot q)}{\partial q_k} = \frac{\partial q_0' L\left(q_{[0]}, \frac{q_{[0]}'}{q_0'}, t\right)}{\partial q_k} = q_0'\frac{\partial L\left(q_{[0]}, \frac{q_{[0]}'}{q_0'}, t\right)}{\partial q_k} = q_0'\frac{\partial L(q_{[0]}, \dot q_{[0]}, t)}{\partial q_k}$$

$$\frac{d}{d\beta}\frac{\partial \mathcal{L}(q,\dot q)}{\partial q'_k}-\frac{\partial \mathcal{L}(q,\dot q)}{\partial q_k}=q'_0\left[\left(\frac{d}{d\beta}\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial \dot q_k}\right)\Big/\left(\frac{dt}{d\beta}\right)-\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial q_k}\right]$$

$$=q'_0\left[\frac{d}{dt}\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial \dot q_k}-\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial q_k}\right]=0$$

此即通常的拉格朗日方程

对 $k=0$，

$$\frac{\partial \mathcal{L}(q,\dot q)}{\partial q'_0}=\frac{\partial q'_0 L\left(q_{[0]},\frac{q'_{[0]}}{q'_0},t\right)}{\partial q'_0}=L\left(q_{[0]},\frac{q'_{[0]}}{q'_0},t\right)+q'_0\frac{\partial L\left(q_{[0]},\frac{q'_{[0]}}{q'_0},t\right)}{\partial q'_0}$$

$$=L(q_{[0]},\dot q_{[0]},t)-\frac{1}{q'_0}\sum_{k=1}^{D}\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial \dot q_k}q'_k=L(q_{[0]},\dot q_{[0]},t)-\sum_{k=1}^{D}\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial \dot q_k}\dot q_k$$

$$=-H(q_{[0]},\dot q_{[0]},t)$$

$$\frac{\partial \mathcal{L}(q,\dot q)}{\partial q_0}=\frac{\partial q'_0 L\left(q_{[0]},\frac{q'_{[0]}}{q'_0},t\right)}{\partial q_0}=q'_0\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial q_0}=\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial t}$$

故方程成为：$\frac{dH(q_{[0]},\dot q_{[0]},t)}{dt}=-\frac{\partial L(q_{[0]},\dot q_{[0]},t)}{\partial t}$ 广义功能定理

$\mathcal{L}(q,\dot q)=q'_0 L\left(q_{[0]},\frac{q'_{[0]}}{q'_0},t\right)$ 为 q'_k 的一次齐次函数，欧拉－拉格朗日方程必有一个不独立。

(11) (10 分)

依题意：$\tilde L(q,\dot q,t)=L(q,\dot q,t)+\frac{df(q,t)}{dt}$ 即：$\tilde L(q,\dot q,t)=L(q,\dot q,t)+\sum_k\frac{\partial f(q,t)}{\partial q_k}\dot q_k+\frac{\partial f(q,t)}{\partial t}$

(a) $\tilde p_k=\frac{\partial \tilde L(q,\dot q,t)}{\partial \dot q_k}=\frac{\partial L(q,\dot q,t)}{\partial \dot q_k}+\frac{\partial f(q,t)}{\partial q_k}$，$\tilde p_k=p_k+\frac{\partial f(q,t)}{\partial q_k}$

(b) $\tilde H(q,\dot q,t)=\sum_k\tilde p_k\dot q_k-\tilde L(q,\dot q,t)$

$$=\sum_k\left[p_k+\frac{\partial f(q,t)}{\partial q_k}\right]\dot q_k-\left[L(q,\dot q,t)+\sum_k\frac{\partial f(q,t)}{\partial q_k}\dot q_k+\frac{\partial f(q,t)}{\partial t}\right]$$

$$=\sum_k p_k\dot q_k-L(q,\dot q,t)-\frac{\partial f(q,t)}{\partial t}=H(q,\dot q,t)-\frac{\partial f(q,t)}{\partial t}$$

$\tilde H(q,\dot q,t)=H(q,\dot q,t)-\frac{\partial f(q,t)}{\partial t}$ 两个广义能量函数之间的关系

(c) 利用 $\dot q_k=\dot q_k(q,p,t)$ 可得：

$$\tilde H(q,p,t)=H(q,p,t)-\frac{\partial f(q,t)}{\partial t}$$

利用 $\tilde p_k=p_k+\frac{\partial f(q,t)}{\partial q_k}$ 消去 p_k，上式可化为

$$\tilde H(q,p',t)=H\left(q,p'-\frac{\partial f(q,t)}{\partial q},t\right)-\frac{\partial f(q,t)}{\partial t}$$

其中 $p'-\frac{\partial f(q,t)}{\partial q}=p$

(12) (9 分)

以空中 x 段为研究主体在 t 时刻，动量为 $m v=(\eta x)\dot x$

在 $t+dt$ 时刻，动量为 $(m+dm)(v+dv)=(\eta x+\eta dx)(\dot x+d\dot x)$

受力的冲量：$(T-\eta g x)\,dt$

从而：$(\eta x+\eta\,dx)(\dot{x}+d\dot{x})-(\eta x)\dot{x}=(T-\eta g x)\,dt$

即：$\eta\dot{x}\,dx+\eta x\,d\dot{x}=(T-\eta g x)\,dt$

对重物 M，有：$M\dfrac{d\dot{X}}{dt}=M g-T$，

消去 T，并利用 $\dot{X}=\dot{x}$， 得

$$\frac{d}{dt}\left[(M+\eta x)\dot{x}\right]=M g-\eta g x$$

同理当 x 减小时： $\eta x\dfrac{d\dot{x}}{dt}=T-\eta g x$

联立重物的方程： $M\dfrac{d\dot{X}}{dt}=M g-T$

得：$(M+\eta x)\dfrac{d}{dt}\dot{x}=M g-\eta g x$